Zonghe Shao

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Education

Huazhong University of Science and Technology, Wuhan, China Sep. 2020-Jul. 2024
Bachelor of Engineering, Artificial Intelligence(expected)

- **Overall GPA:** 3.93/4.0 or 92/100 (rank: top10)
- **Major courses:**

Core Course	Credit	Result	Core Course	Credit	Result
Calculus	5.5	97	Advanced Programming Language C	3.5	95
Linear Algebra	2.5	89	Python Programming	2.5	98
Electronic Technology	5.0	98	Computer Programming(C++)	3.0	97
Computer Networks	2.5	94	Data Structure And Algorithmic Analysis	4.0	95
Digital Signal Processin	2.0	96	Database Technology	2.0	97
Digital Image Processin	2.0	91	Principle Of Automatic Control	3.5	94
Artificial Intelligence	2.0	92	Optimization Theory And Method	3.0	94
Machine Learning	2.0	96	Document Retrieval and Scientific Paper Writing	1.0	95

Academic Research Experience

Multi-task learning based on facial action units Dec. 2022-Present
Research Intern, instructed by A.P. Xiang Xiang

- Read papers about human face expressions recognition and analysis based on action units and proactively keep abreast of the latest developments in this field
- Proposed a method to estimate the intensity of action units in face based on heatmap
- Strove to design an approach for multi-task face analysis including expression recognition and intensity estimation based on the proposed heatmap method

Power system transient stability assessment based on deep learning Feb. 2022-Jan. 2023
Leader, instructed by A.P. Lu Renzhi

- Developed a new feature engineering approach with feature extraction and an improved wrapper approach
- Proposed an LSTM-SAF model for power system transient stability assessment
- Designed a new focal loss function to focuse more on unstable samples with less data
- ***Publication****(in revision):*
  A Novel Data-Driven LSTM-SAF Model for Power System Transient Stability Assessment.(IEEE Transactions on Industrial Informatics, **First author**)

Design of the software named "Ping An Good Car Owner" Dec. 2020-May 2021
Leader, instructed by Prof. Zhou Chunjie and Prof. Gao Changxin

- Constructeed a human-computer interaction software based on C language
- Realized a variety of software functions, such as vehicle registration, insurance purchase and claim settlement
- Constructed a data management method based on the dual mode of user and administrator

Research Interests

GOAL: make AI more perceptive and broaden the real-life application scenarios of AI

- To deeply integrate computer vision technology with medical technology, as an aid to real-world physicians during treatment
- To incorporate 3D scene visual information to make machines more human-like and improve the machine's ability to respond effectively to complex world scenarios
- To build strong perceptive AI using rich information from realistic scenes and respond delicately to the environment

Efforts towards the goals

- Joined the HAIV lab and deep reinforcement learning lab and took the initiative to learn the basics and cutting edge work in the field
- Pondered the problems the current work faced and put forward my views, and actively discussed my views with others
- Proactively verified the feasibility of my ideas and tried to make my innovative work on AI algorithms

Additional Activity

Artificial intelligence elite training program in Baidu Nov. 2022-Present
Remote Member

- Learned and mastered the design and application of AI algorithms based on PaddlePaddle in Baidu
- Provided standard solutions to open algorithm design problems on the platform matiji
- Designed the format and content of the post-class assignments for some chapters in the public machine learning course of Baidu ai studio

Extra-curriculum Activity

Alumni Association and Student Development Guidance Center Oct. 2021-Persent
Core Member

- Planned and organized several alumni homecoming lectures and acted as the venue director
- Arranged and coordinated the schedule of returning alumni in the 70th anniversary of the university and acted as the area manager in the main venue
- Hosted the Light Run on campus with over 10,000 participants

Grants Committee Oct. 2020-Sep. 2022
Directors

- Manage and review students' qualifications for financial aid and scholarships
- Hosted public campaign speeches for various scholarships and honors
- Led the Grants Committee to campus team of the year honors during my tenure as head

Volunteer Association Sep. 2020-Present
Member

- Persisted in volunteering activities and was determined to contribute my efforts in environmental protection and social service.
- Served as a session leader and assisted the moderator in organizing the session agenda at the Artificial Intelligence Conference in China Optics Valley

Awards and Honors

Second Prize in the Blue Bridge Cup Algorithm Design Competition	Apr. 2023
Second Prize in Mathercup Mathematical Modeling Challenge	May 2022
Meritorious Winner in Mathematical Contest in Modeling	Feb. 2022
Second Prize in Chinese College Students Mathematics Contest	Nov. 2021
Second prize in the mathematics competition for college students in Hubei Province	Oct. 2021
National Encouragement Scholarship (Twice)	Sep. 2022 & Sep. 2021
Study Excellence Scholarship (Twice)	Sep. 2022 & Sep. 2021
Outstanding Freshman Award	Jan. 2021

Skills and Interests

- **Languages**: English (fluent, **academic writing**) CET-4: 587 CET-6: 511
- **Programming Language**: C/C++, Python, MATLAB, Latex, SQL
- **DL Framework**: Pytorch, Tensorflow, PaddlePaddle
- **Amateur Interests**: table tennis, badminton, yoga